INCH-POUND

MS27477J
w/AMENDMENT 1
10 September 2024
SUPERSEDING
MS27477J
13 April 2012

DETAIL SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, JAM NUT MOUNTING, SOLDER TYPE, HERMETIC SEAL, SERIES II

Reactivated after 24 May 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

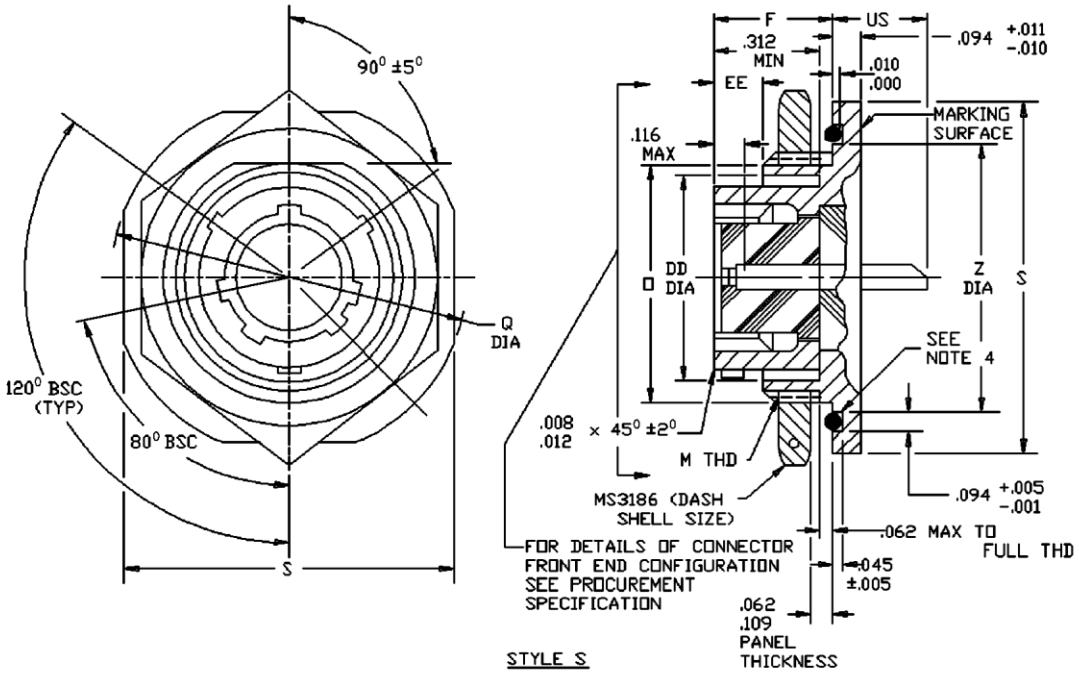


FIGURE 1. Receptacle, class Y.

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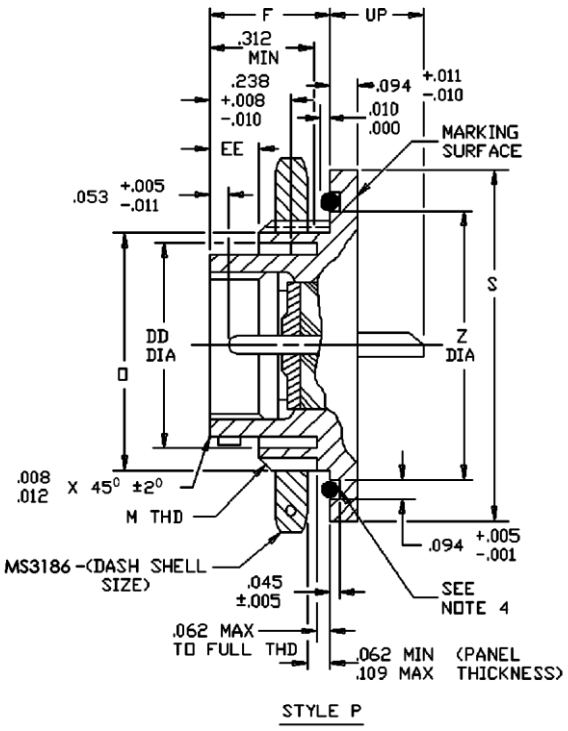


FIGURE 1. Receptacle, class Y – Continued.

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Shell size	F ± .005	M Thread, class 2A (plated)	O + .001 -.006 flat	Q dia ±.016	S ±.016
8	.438	.875 – 20 UNEF	.817	1.375	1.250
10		1.000 – 20 UNEF	.941	1.500	1.375
12		1.125 – 18 UNEF	1.065	1.625	1.500
14		1.250 – 18 UNEF	1.190	1.750	1.625
16		1.375 – 18 UNEF	1.320	1.938	1.781
18		1.500 – 18 UNEF	1.440	2.016	1.890
20	.464	1.625 – 18 UNEF	1.565	2.141	2.016
22		1.750 – 18 UNS	1.690	2.265	2.140
24		1.875 – 16 UN	1.815	2.390	2.265

Shell size	UP max	US max	Z +.006 -.005 dia	DD min dia	EE +.005 -.000	Recommended Packing (O-ring) (see SAE-AS3578) (see note 7)
8	.281	.359	.965	.678	.140	AS3578-022
10			1.090	.780		AS3578–024
12			1.215	.963		AS3578–026
14			1.340	1.088		AS3578–028
16			1.477	1.222		AS3578–029
18			1.602	1.333		AS3578–030
20	.250	.344	1.727	1.458	.166	AS3578–031
22			1.852	1.583		AS3578–032
24			1.977	1.708		AS3578–033

FIGURE 1. Receptacle, class Y – Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.250	6.35	1.190	30.23	1.727	43.87
.003	0.08	.281	7.13	1.215	30.86	1.750	44.45
.005	0.13	.312	7.92	1.222	31.04	1.781	45.24
.006	0.15	.344	8.74	1.250	31.75	1.815	46.10
.008	0.20	.359	9.12	1.320	33.53	1.852	47.04
.010	0.25	.438	11.13	1.333	33.86	1.875	47.63
.011	0.28	.464	11.79	1.340	34.04	1.890	48.00
.012	0.30	.678	17.22	1.375	34.95	1.938	49.23
.016	0.41	.780	19.81	1.440	36.58	1.977	50.22
.045	1.14	.817	20.75	1.458	37.03	2.016	51.21
.053	1.35	.875	22.23	1.477	37.52	2.140	54.36
.062	1.57	.941	23.90	1.500	38.10	2.141	54.38
.094	2.39	.963	24.46	1.565	39.75	2.265	57.53
.109	2.77	1.000	25.40	1.583	40.21	2.390	60.71
.116	2.95	1.065	27.05	1.602	40.69		
.140	3.56	1.088	27.64	1.625	41.28		
.166	4.22	1.090	17.69	1.690	42.93		
.238	6.05	1.125	28.58	1.708	43.38		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Jam nut mounting receptacles shall be provided with an O-ring.
4. Sides of groove may have 5° maximum taper.
5. For main key polarization and key positions, see MIL-DTL-38999.
6. Diameters M, Q and Z shall be concentric within .015 inches (0.381 mm) T.I.R.
7. Recommended packing is for dimensions only, material is optional.

FIGURE 1. Receptacle, class Y – Continued.

REQUIREMENTS:

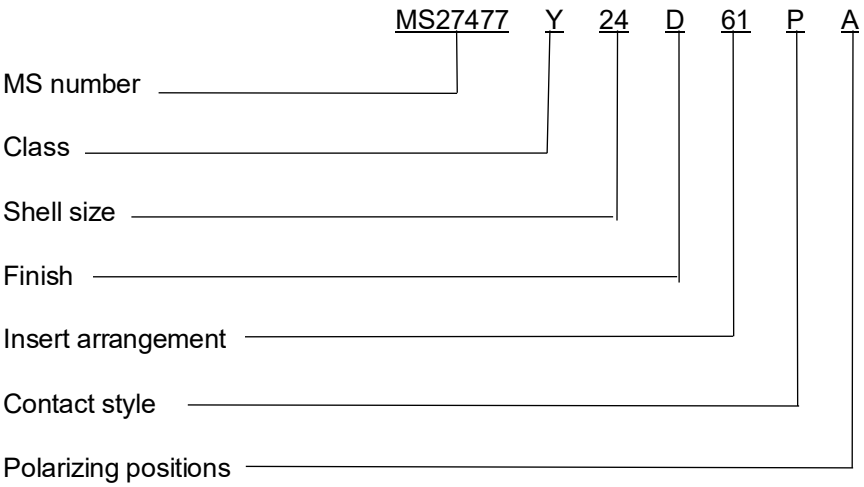
Dimensions and configuration: See figure 1. Interface dimensions shall conform to MIL-DTL-38999. This receptacle mates with MS27473, MS27480, MS27484 and MS27500.

Insert arrangements: See MIL-STD-1560.

Hex nuts: See MS3186.

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Part or Identifying Number (PIN) example:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-STD-1560
MS3186
MS27473
MS27480
MS27484
MS27500
SAE-AS3578

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force - 85
DLA – CC

Preparing activity:
DLA – CC

(Project: 5935-2024-163)

Review activities:
Army – MI
Navy – EC, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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